

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 06 — INTRODUCTION TO MODBUS

Student Edition

Authorized ICDFA training environments only
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Lab 06: Introduction to Modbus

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 3–4 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the difference between Modbus TCP and Modbus RTU.
- Identify common Modbus function codes and data areas.
- Observe Modbus TCP communication between FUXA and OpenPLC in an isolated lab.
- Use Wireshark to capture/filter Modbus traffic and identify request/response fields.
- Relate register/address information to the running tank-fill process.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password
SCADA VM	192.168.2.10	scada	password
Analyst/Kali VM	Instructor-provided	kali	kali

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Reference: Modbus Function Codes

Code	Function	Purpose
01	Read Coils	Read discrete output/coils
02	Read Discrete Inputs	Read discrete input states
03	Read Holding Registers	Read 16-bit holding registers
04	Read Input Registers	Read 16-bit input registers
05	Write Single Coil	Change one discrete coil
06	Write Single Holding Register	Change one holding register
15	Write Multiple Coils	Change multiple coils
16	Write Multiple Holding Registers	Change multiple registers

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Start the PLC, SCADA and analyst VMs on the instructor-provided isolated laboratory topology. Do not connect this lab to production or bridged networks.

Step 2. On the PLC VM, start the TankFillSim program in OpenPLC Runtime and confirm Running status.

Step 3. On the SCADA VM, start FUXA, load the tank-fill HMI and confirm the openplc connection targets 192.168.1.10:502.

Step 4. Run the HMI and verify that the tank level changes between its operating limits.

Step 5. On the analyst VM, open Wireshark. Select only the interface connected to the ICDFA lab segment carrying PLC/SCADA traffic.

Step 6. Apply the display filter modbus. If needed, refine it with tcp.port == 502.

Step 7. Start a short packet capture while the HMI is running, allow several process updates to occur, then stop the capture.

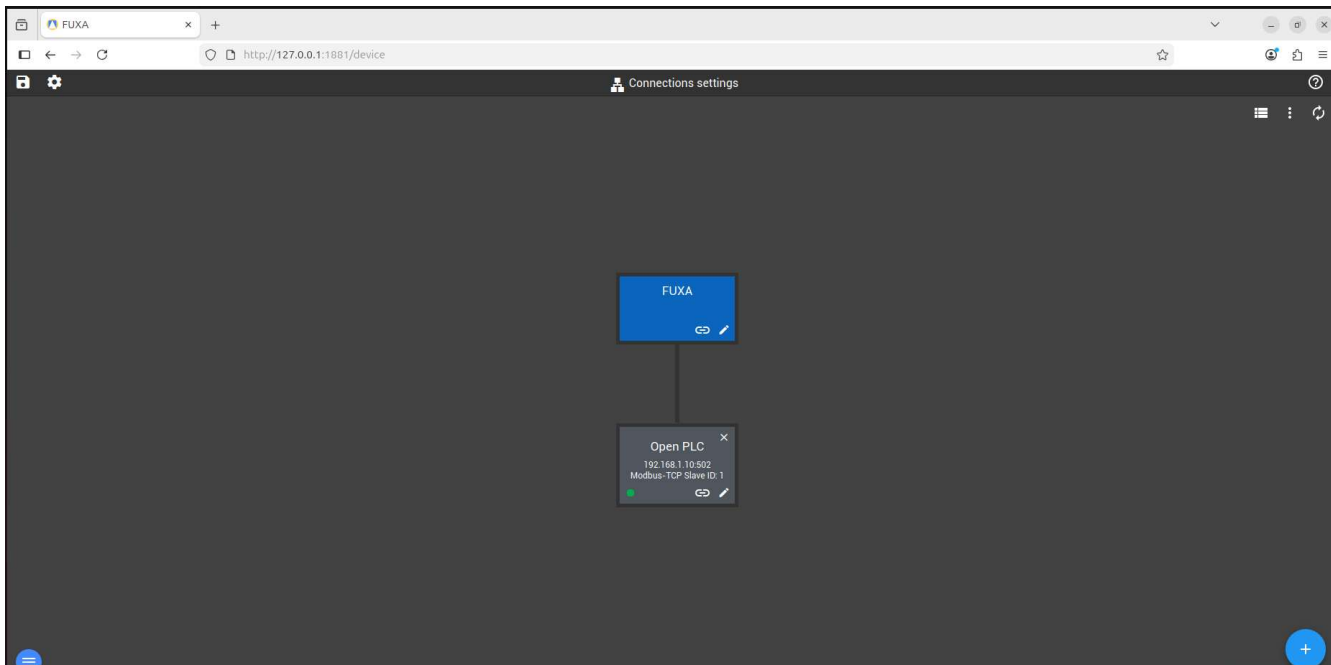
Step 8. Select a Modbus request/response pair and expand the Modbus/TCP and Modbus protocol fields.

Step 9. Identify the transaction/request, source and destination IP addresses, function code and referenced register/data value.

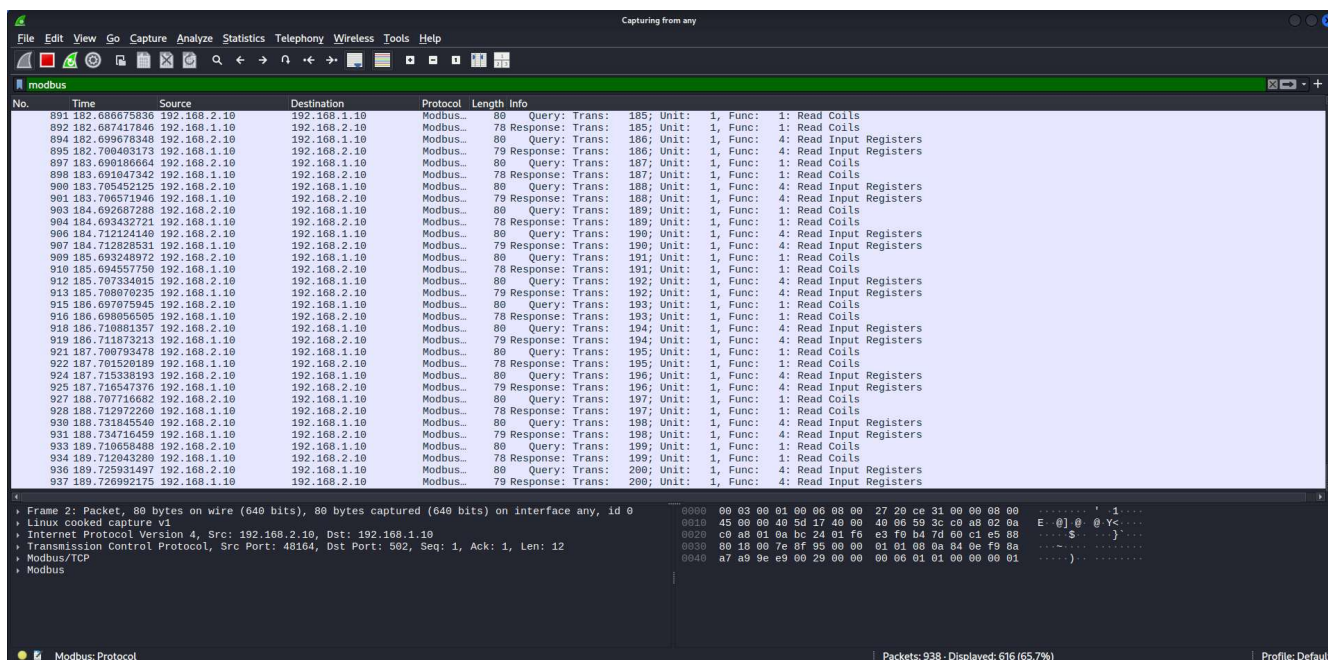
Step 10. Locate a Function Code 4 (Read Input Registers) response associated with the tank-level value. In the teaching image, Input Register 80 corresponds to the Tank Level indication.

Step 11. Complete the Modbus function-code table by explaining at least codes 01, 02, 03, 04, 05, 06, 15 and 16.

Step 12. Save the packet capture with a filename containing your registration number and LAB06. Do not capture unrelated traffic.

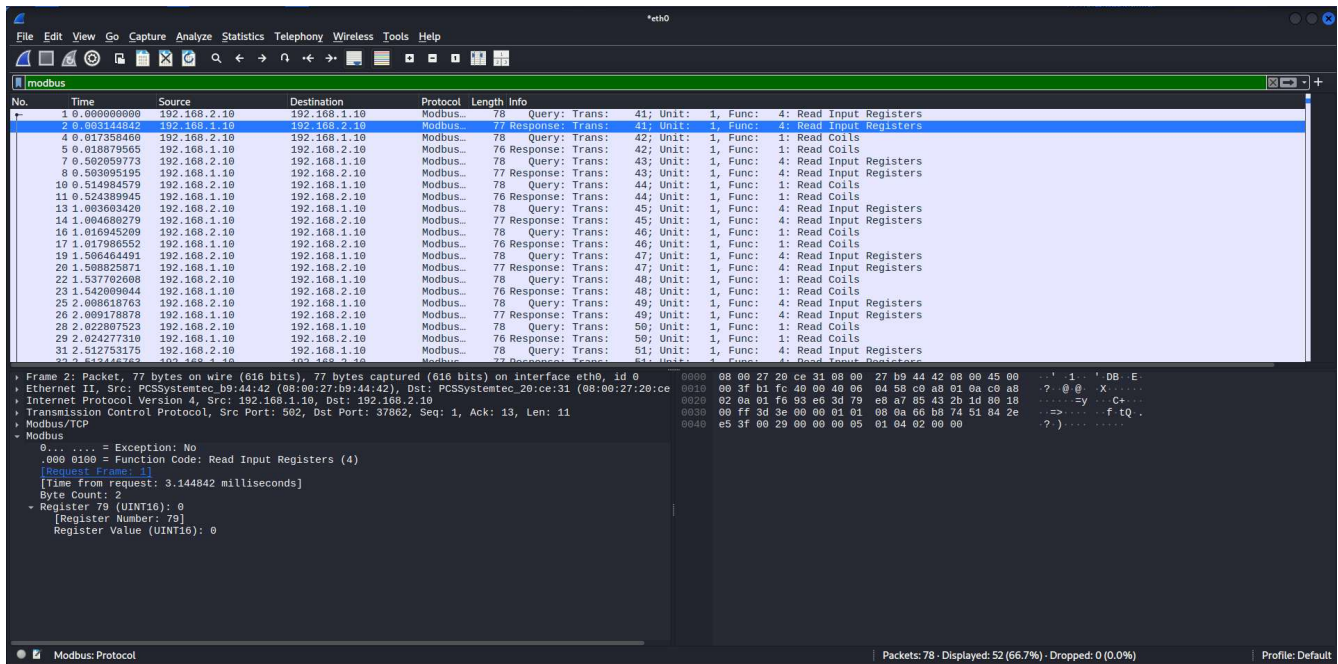


IMG: FUXA/OpenPLC process running.



IMG: Wireshark filtered to Modbus traffic.

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The image is a specific Response Query, it shows a Response from Register 80 which can be found by expanding the Modbus entry. This is the *Tank Level* data, which in this case, is reading 0ft. This Modbus Response is going from the *OpenPLC Runtime* PLC simulation to the *FUXA SCADA* system.

Function Code Table

Function	Code	Bit Size
Read Single Coil	01	1
Read Discrete Inputs	02	1
Read Multiple Holding Registers	03	16
Read Input Registers	04	16
Write Single Coil	05	1
Write Single Holding Register	06	1
Write Multiple Coils	15	16
Write Multiple Holding Registers	16	16

Modbus TCP versus Modbus RTU

Modbus RTU is the typical method for serial communication, it transmits binary data one bit at a time. Serial networks are used in industrial applications where the data needed is not time critical. Modbus RTU is still an option on many industrial devices due to its prevalence in many industrial systems. It uses a primary-secondary architecture. Modbus TCP/IP is the standard for parallel communication based on ethernet. It is typically used in modern industrial applications and those in which the speed of data transmission is critical to the process. It uses a client-server topology.

Required Evidence & Submission

- Screenshot of the FUXA/OpenPLC process running.
- Screenshot of Wireshark filtered to Modbus traffic.
- One annotated request/response analysis identifying function code and register.
- The saved .pcap/.pcapng capture from the isolated training network.
- A completed function-code table and a short comparison of Modbus TCP versus Modbus RTU.

Recommended filename: <RegistrationNo>_LAB06_Introduction_to_Modbus.pdf plus the required project/capture file(s).

Knowledge Check

1. What transport/networking difference separates Modbus TCP from Modbus RTU?

Response: Modbus TCP relies on standard Ethernet and TCP/IP networks (typically using TCP port 502) to transport application data frames inside MBAP headers, whereas Modbus RTU uses serial communication lines (like RS-485 or RS-232) with direct binary framing and cyclic redundancy checks (CRC).

2. Why does Modbus use function codes?

Response: When a client polls a server (PLC) for a read/write request, the client needs to know what specific object data, for example, coils, inputs, contacts, etc., to access and what action it needs to take on that data on its end. To ensure the HMI is reading or writing to the PLC about the correct object, function codes are used to define what action to take on the data.

3. What does Function Code 4 request/read?

Response: Function code 4 read input Registers (It reads physical analog sensor inputs).

4. Why is visibility of unprotected industrial protocol traffic a cybersecurity concern?

Response: Unprotected industrial protocol traffic lacks basic encryption and authentication, meaning anyone on the network can passively eavesdrop on sensitive operational data. This unencrypted visibility allows attackers to easily conduct process reconnaissance, map out control logic, and intercept values to craft spoofed commands or execute man-in-the-middle attacks that directly disrupt physical operations.

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.
Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.		